Enhancing the spatial resolution of monthly gravity solutions derived from satellite laser ranging

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Motivation

Why observe monthly gravity variations?

- Changes in the Earth's gravity field reveal the **redistribution of mass** across the Earth system
- NASA and DLR satellites (GRACE) provide global monthly observations used to monitor **hydrology, ice-sheet mass change, and ocean dynamics**
- Long and continuous gravity records are essential for identifying **seasonal, interannual, and long-term changes**

Why is this challenging?

- GRACE observations are only available from 2002 onward and contain several **data gaps**
- Satellite Laser Ranging (SLR) provides a much longer record, but resolves only the **largest spatial scales**
- Recovering smaller-scale gravity variations from SLR is therefore a strongly **underdetermined reconstruction problem**

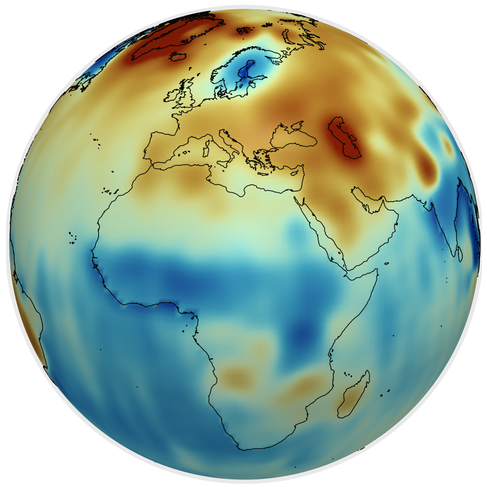


Figure 1. Example of a monthly gravity field reconstruction.

Gravity-Field Representation

The gravity-field anomaly is expanded in spherical harmonics, where ℓ denotes the **degree** and m the **order**. The degree controls the spatial scale of the signal, while the order controls its longitudinal variation:

$$\Delta V(r,\theta,\phi,t) = \frac{GM}{r}\sum_{\ell=2}^{L}\left(\frac{R}{r}\right)^{\ell}\sum_{m=0}^{\ell}\underbrace{\bar{P}_{\ell m}(\cos\theta)}_{\text{fully normalized associated Legendre function}}\left[\Delta\bar{C}_{\ell m}(t)\cos(m\phi)+\Delta\bar{S}_{\ell m}(t)\sin(m\phi)\right].$$

Gravity anomalies are expressed as **equivalent water height (EWH)**: the thickness of a hypothetical water layer producing the same mass anomaly.

Spatial enhancement

Increasing the maximum degree from $\ell = 10$ to $\ell = 60$ substantially refines the spatial resolution:

$$\lambda_\ell \approx \frac{20\,000\ \text{km}}{\ell}.$$

$$\lambda_{10} \approx 2000\ \text{km}, \qquad \lambda_{60} \approx 333\ \text{km}.$$

Spectral coefficient layout

$$S_{60,60}\cdots\ S_{60,1}\,C_{60,0}\,C_{60,1}\ \cdots\ C_{60,60}$$

$$\vdots$$

$$S_{22}\ \ S_{21}\ \ C_{20}\ \ C_{21}\ \ C_{22}$$

$$S_{11}\ \ C_{10}\ \ C_{11}$$

$$C_{00}$$

Reconstruction Definition

$$\mathbf{x}_t = \left\{\Delta\bar{C}^{\text{SLR}}_{\ell m}, \Delta\bar{S}^{\text{SLR}}_{\ell m}\right\}_{\ell\le 10}, \qquad \mathbf{y}_t = \left\{\Delta\bar{C}^{\text{GRACE}}_{\ell m}, \Delta\bar{S}^{\text{GRACE}}_{\ell m}\right\}_{\ell\le 60},$$

$$\widehat{\mathbf{y}}_t = f_\theta(\mathbf{x}_t, \mathbf{c}_t), \qquad \theta^\star = \arg\min_\theta \sum_{t\in\mathcal{T}_{\text{train}}} \mathcal{L}\left(f_\theta(\mathbf{x}_t,\mathbf{c}_t),\mathbf{y}_t\right),$$

Because $\mathbf{x}_t$ does not uniquely determine $\mathbf{y}_t$, the network learns a conditional GRACE-era prior rather than an exact physical inversion.

Data and Preprocessing

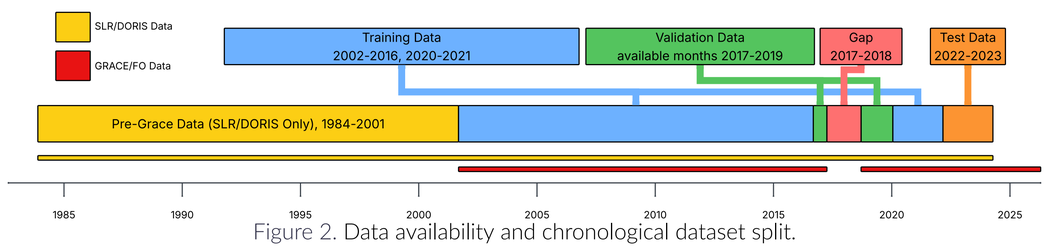


Figure 2. Data availability and chronological dataset split.

Input

Monthly **SLR/DORIS** coefficients.[1]

Target

Monthly **COST-G GRACE/GRACE-FO** combined solutions.[2]

Noise Reduction

A **250 km Gaussian filter** is applied to reduce high degree noise from target data.

Spectrally Structured Training Objective

The final objective minimizes $\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_1^{\text{high}} + 0.01\,\mathcal{L}_1^{\text{low}} + \mathcal{L}_{\text{HDLO}}}$

Notation: B is the mini-batch size, while σ_ℓ is the degree-dependent Gaussian width controlling the weight decrease with order m.

Degree-Balanced Coefficient Loss

For degree ℓ, the set of valid coefficients is

$$\mathcal{I}_\ell = \{(\ell,m,C) : 0\le m\le\ell\}\cup\{(\ell,m,S) : 1\le m\le\ell\}.$$

The mean absolute error of one degree is

$$\mathcal{L}_1^{(\ell)} = \frac{1}{B\,|\mathcal{I}_\ell|}\sum_{b=1}^{B}\sum_{i\in\mathcal{I}_\ell}|\widehat{y}_{b,i} - y_{b,i}|.$$

The generated high degrees are supervised by

$$\mathcal{L}_1^{\text{high}} = \frac{1}{50}\sum_{\ell=11}^{60}\mathcal{L}_1^{(\ell)},$$

while the low-degree residual head is supervised by

$$\mathcal{L}_1^{\text{low}} = \frac{1}{9}\sum_{\ell=2}^{10}\mathcal{L}_1^{(\ell)}.$$

The non-normalized $\mathcal{L}_1$ loss gives greater influence to higher-magnitude coefficients, emphasizing the dominant gravity signals.

High-Degree Low-Order Loss

The order-dependent Gaussian weight is

$$w_{\ell m} = \exp\left(-\frac{m^2}{2\sigma_\ell^2}\right).$$

The reference power of each sample and degree is

$$Q_{b,\ell} = \frac{1}{\ell+1}\sum_{m=0}^{\ell}(y_{b,\ell m})^2.$$

The total Gaussian weight is

$$W = \sum_{\ell=30}^{60}\sum_{m=0}^{\ell} w_{\ell m}.$$

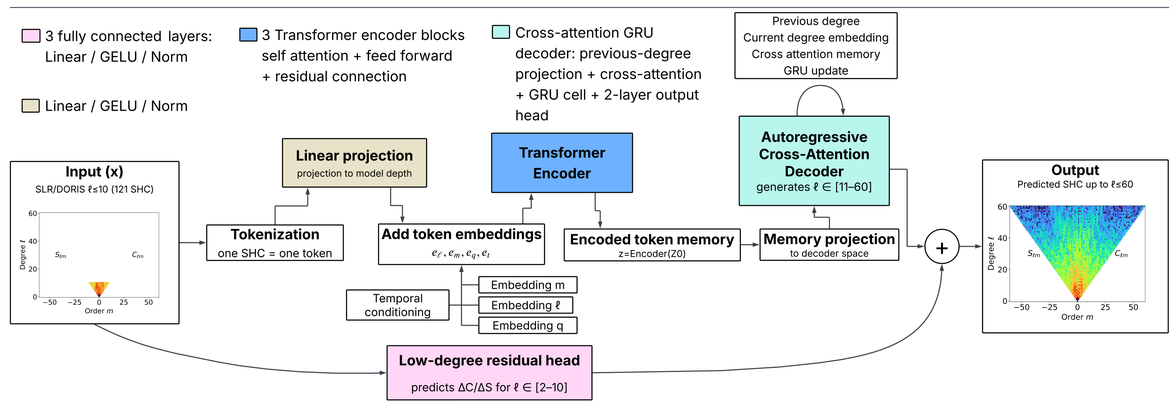
The degree-normalized HDLO loss is

$$\mathcal{L}_{\text{HDLO}} = \frac{\displaystyle\sum_{b=1}^{B}\sum_{\ell=30}^{60}\sum_{m=0}^{\ell} w_{\ell m}\frac{(\widehat{y}_{b,\ell m} - y_{b,\ell m})^2}{Q_{b,\ell}}}{2BW}.$$

Normalization by $Q_{b,\ell}$ makes the loss sensitive to **relative error**, while $w_{\ell m}$ concentrates supervision on the low-order region of the high-degree spectrum.

Optimization principle: broad coefficient accuracy over all reconstructed degrees, reinforced where high-degree low-order errors contribute most strongly to the remaining spatial error.

Degree-Autoregressive Reconstruction Model

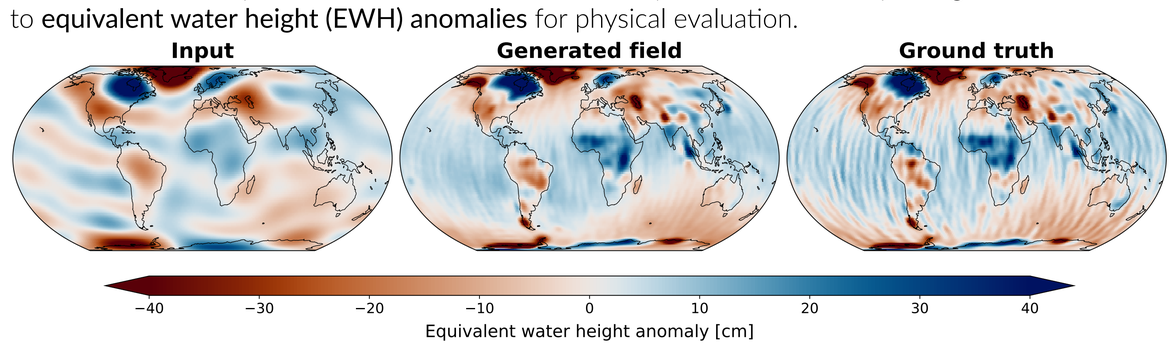


Each observed coefficient is represented as a spectral token. A Transformer encoder combines coefficient structure and temporal context, while a cross-attention GRU decoder generates degrees $11 \le \ell \le 60$ sequentially. A residual head corrects the observed degrees $2 \le \ell \le 10$.

121 input tokens · 3 Transformer blocks, 8 heads · 50 generated degrees · 3,600 predicted coefficients · 3.4 million parameters

Results

The reconstructed spherical-harmonic coefficients are synthesized onto a spatial grid and converted to **equivalent water height (EWH) anomalies** for physical evaluation.



Example of a generated field from test set (December 2022).

Held-out 2022–2023 performance

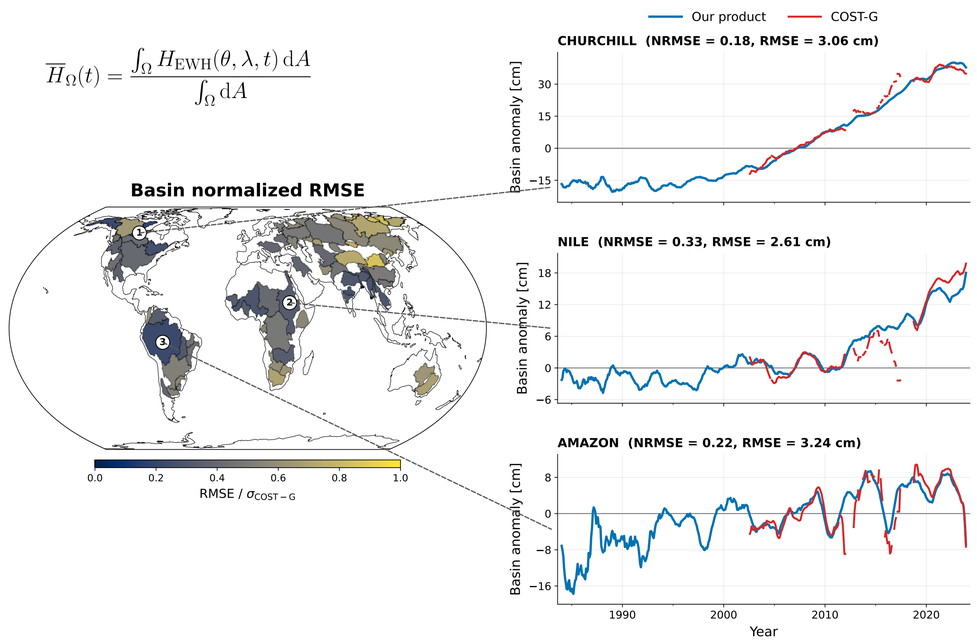
Grid RMSE	Spatial correlation	EDNRMSE	Log-spectrum RMSE
4.06 cm	0.986	0.377	0.062

28.4% **lower grid RMSE than Löcher et al. (2025)** [1] · best performance on 7/8 principal metrics.

Long-Term Application: Hydrology, 1984–2023

Hydrological variability is evaluated over 72 large river basins ($> 200\,000\ \text{km}^2$) using the basin-mean equivalent-water-height anomaly:

$$\overline{H}_\Omega(t) = \frac{\int_\Omega H_{\text{EWH}}(\theta,\lambda,t)\,\mathrm{d}A}{\int_\Omega \mathrm{d}A}$$



GRACE-Era Agreement

Against COST-G over 2002–2023, the reconstruction reproduces basin-scale variability with an area-weighted correlation of $\mathbf{0.883}$ and RMSE of $\mathbf{2.97\ cm}$.

Pre-GRACE Interpretation

The reconstructed series extends basin-scale variability back to 1984, but before 2002 the high-degree contribution remains a **conditional GRACE-era reconstruction**, not an independent observation.

Conclusion

Main Contributions

- Developed a degree-autoregressive model that reconstructs degree-60 gravity fields from degree-10 SLR/DORIS observations.
- Introduced a spectrally structured objective.
- Improved spatial, spectral, and basin-scale performance relative to the existing SLR/DORIS reconstruction benchmark.
- Extended the reconstruction across the 1984–2024 SLR/DORIS record.

Limitations and Outlook

- High-degree information is inferred from statistical relationships learned during the GRACE era rather than directly observed by SLR/DORIS.
- Reconstruction accuracy decreases toward degree $\ell = 60$ and degrades for temporally out-of-distribution states.

Take-Home Message

Deep learning can transfer spatial information learned during the GRACE era to long-term, low-resolution SLR/DORIS gravity solutions, producing more informative monthly mass fields while preserving their extended temporal coverage.

References

[1] Anno Löcher, Jürgen Kusche, and Yufeng Nie. A 40-year record of the earth's time-variable gravity field from slr and doris. *Advances in Space Research*, 76(3):1281–1291, 2025.

[2] Adrian Jäggi et al. International combination service for time-variable gravity fields (COST-G). In *International Association of Geodesy Symposia*. Springer, 2020.